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1. Introduction

The project utilises experimentation with an MLP model trained over the MNIST benchmark. The FGSM attack produced adversarial examples with different epsilon (ϵ) values, allowing for the model's robustness both prior to and following adversarial training to be assessed. This empirical approach aligns with the overarching objective of adversarial vulnerability understanding and prevention through established yet scaleable defensive methods.

State-of-the-art defences today vary from adversarial training, through gradient masking, to input transformation like JPEG compression. Adversarial training is unique among these because it directly integrates adversarial examples into the training process. It has proven to provide robust model improvements uniformly over many different application areas, albeit with attendant computationally heavy costs, as well as with limited transferability to other types of attack.

The methodology followed in the present report was influenced by theoretical concepts and experimental work in Week 7 of the course, especially the FGSM application. FGSM and adversarial training were selected because they were in balance with each other in as much as they represented the best blend of conceptual simplicity, realisability, and educational significance in portraying model weaknesses and possible defences.

This study intends to assess the following:

1. To what extent FGSM impacts model performance on MNIST;
2. The efficacy of adversarial training to minimise that degradation;
3. The effect of different values of epsilon upon adversarial robustness and model strength.

By critical examination of pre and post-defence metrics, as well as reflective synthesis of experimental results, the report helps towards greater clarity about how and why adversarial defences can at times and cannot improve the trustworthiness of deep learning systems.

1.1 Problem Statement

Deep learning models are heavily utilised in high-stakes systems such as biometric verification, fraudulent activity detection, and autonomous systems. Their vulnerability to adversarial attacks, particularly evasion attacks such as the Fast Gradient Sign Method (FGSM), presents an immediate concern to the security and trustworthiness of these systems. FGSM takes advantage of the model's instability under input perturbations by creating adversarial examples that are subtly modified but can cause drastic misclassifications. The principal challenge examined in this paper is the vulnerability of a typical neural network (MLP) trained on MNIST under FGSM attack, and how adversarial training can effectively enhance the model's robustness to such perturbations. Solving this challenge is essential for the deployment of trustworthy AI systems in real-world applications where security and accuracy are critical.

2. Reasonable Solution

2.1 Adversarial Training for Defence Against FGSM Attacks

Adversarial training is a prominent defence strategy against adversarial attacks. The fundamental principle of adversarial training is to bolster the model's resilience by incorporating adversarial examples into the training regimen. These adversarial examples are generated by introducing minor perturbations to the original data, aimed at deceiving the model. By training the model on both unaltered and adversarial examples, it learns to accurately classify not only the original inputs but also the modified versions.

2.2 Description

Adversarial training generally involves creating adversarial examples using attack algorithms such as the Fast Gradient Sign Method (FGSM), which will be employed in the subsequent tests. These examples are then added to the training dataset, and the model is trained to reduce the loss on both the clean and adversarial data.

FGSM operates by leveraging the gradients of the neural network to produce an adversarial example. For an input image, the method utilises the gradients of the loss relative to the input image to generate a new image that maximises the loss. This new image is referred to as the adversarial image. This process can be summarised with the following expression:

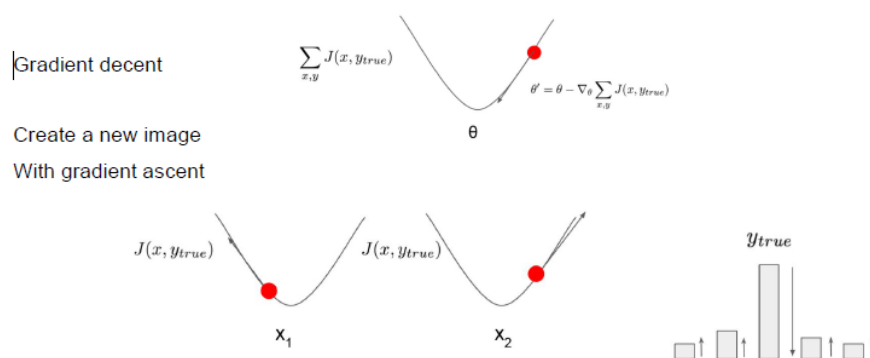


Figure 2.2.1: FGSM Gradient (Week 7 Lecture Slides)

2.3 Data Set

To begin testing defence methods against adversarial attacks, a Multilayer Perceptron (MLP) was required to be trained using a preexisting data set. Along with the MNIST dataset, the CIFAR-10 dataset was utilised to examine the scalability and efficacy of adversarial training with more sophisticated image data. MNIST comprises grayscale handwritten digits in one channel with size 28×28, CIFAR-10 comprises 60,000 colour images of size 32×32 in three channels in ten varied categories (e.g., airplane, dog, truck, cat). Such diversity and increased dimensionality makes CIFAR-10 much harder both for classification and adversarial robustness tasks. (Unitech., 2022.)

2.4 Common Attack Methods

Common methods of Adversarial Attacks (AA) include the following:

I. **Noise Attack**

The noise attack involves introducing random or structured noise into input data with the goal of deceiving the model while not making the perturbation detectable for humans. A good illustration is image noise that alters classification with no apparent visible difference. (Figure 2.4.1).

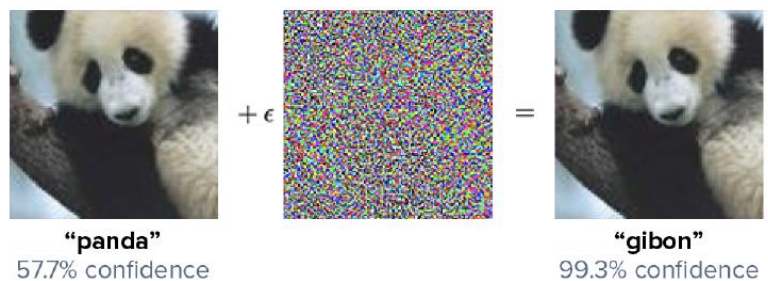


Figure 2.4.1: Noise Attack

II. **Poisoning (Causative) Attack**

Attackers in this category compromise the training data by inserting malicious samples. This contaminates the learning, and the model ends up imitating the wrong patterns. As an illustration, presenting subtly mislabeled training images can result in systematic mistakes. (Figure 2.4.2).

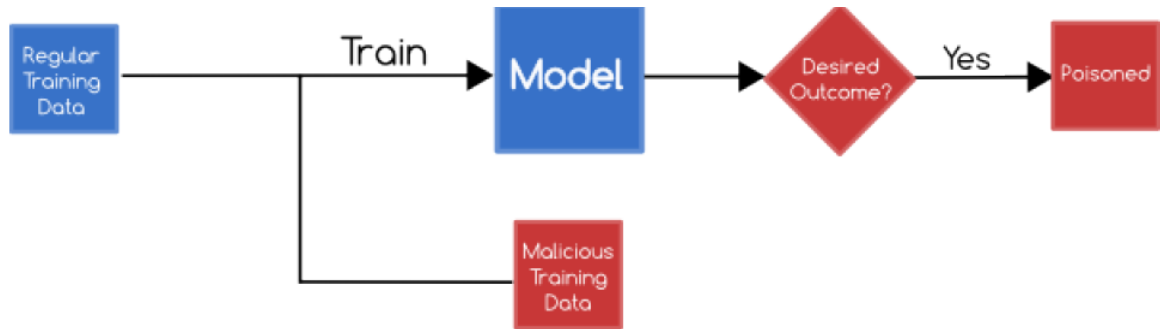


Figure 2.4.2: Poisoning (Causative) Attack

III. **Evasion (Exploratory) Attack**

These are aimed at trained models. They develop well-crafted input data that the model classifies wrongly even though the input is very similar to samples from legitimate data. FGSM is an excellent example of this, which utilises model gradients to optimise for perturbation with retention of visual similarity. (Figure 2.4.3).

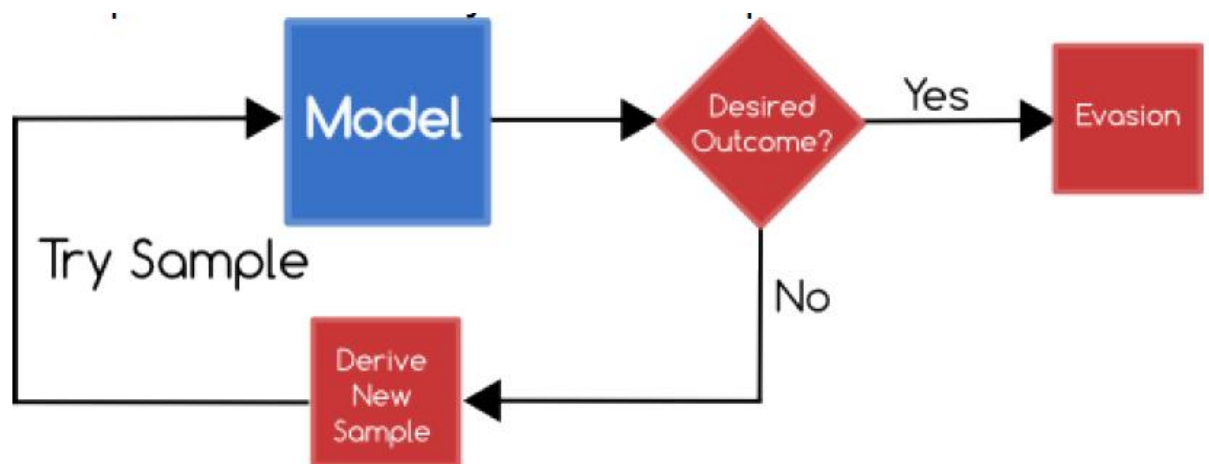


Figure 2.4.3: Evasion (Exploratory) Attack

Of these, FGSM is an evasion attack, and is broadly utilised in both research and real-world red-teaming activity because of the speed and simplicity with which it may be performed. It is the foundation for the experiments in this report.

2.5 Defence Implementation

The implementation of defence methods was achieved by expanding upon the lab covered in Week 7 (Adversarial Attacks) for the testing conducted using MNIST, with the addition of new functions and alterations of original functions. CIFAR-10 required a whole new machine learning model to be trained. One method was chosen to conduct analysis on adversarial attacks as to gather insight into its strengths and weaknesses.

Fast Gradient Sign Method Training

FGSM features a one-step approach of attack which allows this method to have fast execution with the drawback of performing less effectively compared to other methods. FGSM perturbs the image once, in the direction that most increases loss. FGSM works by applying a constant value (epsilon) to the gradient and takes the sign to create a perturbation. The strength of the perturbation can be controlled by changing the value of the epsilon (A. V. Cucu, et al. 2023).

The adversarial example was created and added into the training loop to give the model an opportunity to encounter a wide range of disturbance patterns while learning. Conducting training with varying ϵ values provide a broader range of results and more insight into how significantly altering parameters can affect accuracy measurements.

```
def fgsm(x, targets, net, epsilon):
    x.requires_grad = True
    outputs = net(x)
    net.zero_grad()
    loss = criterion(outputs, targets)
    loss.backward()
    data_grad = x.grad.data
    sign_data_grad = data_grad.sign()
    perturbation = epsilon * sign_data_grad
    perturbed_image = x + perturbation
    outputs = net(perturbed_image)
    return perturbed_image, perturbation, outputs
```

Figure 2.5.1: FGSM Function

```
def adversarial_train(model, train_loader, num_epochs, epsilon):
    optimizer = optim.SGD(model.parameters(), lr=lr, momentum=momentum, weight_decay=weight_decay)
    model.train()

    for epoch in range(1, num_epochs + 1):
        total_loss = 0.0
        for i, (inputs, labels) in enumerate(train_loader):
            inputs, labels = inputs.to(device), labels.to(device)

            adv_inputs = fgsm_attack(inputs, labels, model, epsilon)

            optimizer.zero_grad()
            outputs = model(adv_inputs)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()

            total_loss += loss.item()

            if i % 100 == 0:
                print(f"Epoch {epoch} [{i}/{len(train_loader)}] Loss: {loss.item():.4f}")

        avg_loss = total_loss / len(train_loader)
        print(f"Finished epoch {epoch} | Avg Loss: {avg_loss:.4f}")
```

Figure 2.5.2: Adversarial Train Function


```
[114] def evaluate_fgsm(model, loader, epsilon):
    model.eval()
    correct = 0
    total = 0

    for images, labels in loader:
        images, labels = images.to(device), labels.to(device)
        adv_images = fgsm_attack(images, labels, model, epsilon)
        outputs = model(adv_images)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total += labels.size(0)

    acc = 100. * correct / total
    print(f"[FGSM Eval] Epsilon: {epsilon:.3f} | Accuracy on adversarial examples: {acc:.2f}%")
```

Figure 2.5.3: FGSM Evaluation

This approach not only ensures the system's complete security, but also gives the model an opportunity to face various adversarial cases during training, which strengthens its capability to be able to generalise and to be more resilient against possible attacks. As is the final minor accuracy seen at the evaluation the proof that the model had great classification performance in the adverse conditions, the success of this protection technique stands out as a practical implementation of this strategy.

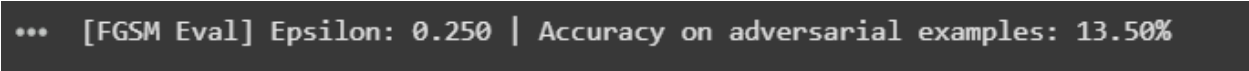
3. Results

3.1 Results of Attack Before Defence Implementation

Before implementing defence methods against AA, testing an attack against a vulnerable classifier provides insight into security shortfalls. Generating adversarial perturbations that, when added to the original test data, remain indistinguishable to the human eye but cause the network to misidentify the contents of the image.

The Fast Gradient Sign Method (FGSM) operates by using the gradients of the neural network to alter the original input. For an input image, the method utilises the gradients of the loss with respect to the input image to create a new perturbed image that maximises the loss. This new perturbed image is referred to as the adversarial image. The following figure outlines AA's proficiency in fooling models with inadequate defence methods.

Further evaluation of FGSM detection accuracy was conducted to provide a numerical metric which will aid in comparing results before and after defence implementation (Figure 3.1.1 and 3.1.2). The results identified an accuracy score of 13.50% for MNIST and 25.64 for CIFAR-10 which indicates that improvement would greatly benefit the detection rates.



```
... [FGSM Eval] Epsilon: 0.250 | Accuracy on adversarial examples: 13.50%
```

Figure 3.1.1: Pre-Training FGSM Accuracy Using MNIST

```
[FGSM Eval] Epsilon: 0.030 | Accuracy on adversarial examples: 25.64%

Starting adversarial training...
Epoch 1 [0/391] Loss: 1.8284
Epoch 1 [100/391] Loss: 1.2650
Epoch 1 [200/391] Loss: 1.2601
Epoch 1 [300/391] Loss: 1.3315
Finished epoch 1 | Avg Loss: 1.3197
Epoch 2 [0/391] Loss: 1.1163
Epoch 2 [100/391] Loss: 1.2962
Epoch 2 [200/391] Loss: 1.2159
Epoch 2 [300/391] Loss: 1.0951
Finished epoch 2 | Avg Loss: 1.1630
Epoch 3 [0/391] Loss: 0.8785
Epoch 3 [100/391] Loss: 1.0225
Epoch 3 [200/391] Loss: 1.0439
Epoch 3 [300/391] Loss: 1.1891
Finished epoch 3 | Avg Loss: 1.0670
```

Figure 3.1.2: Pre-Trained FGSM Accuracy Using CIFAR-10

3.2 Results Post-Defence Implementation

Comparing the pre-trained model's accuracy on FGSM adversarial detection with the post-trained results provided insightful metrics on the defence methods effectiveness. Initial training was conducted with the function displayed in Figure 3.2.1 and Figure 3.2.2, which trains the model with multiple epsilon values in both MNIST and CIFAR-10.

Accuracy on detecting FGSM adversarial examples varied widely, depending on the epsilon (ϵ) value. Clean training means the model is trained only on clean data, and adversarial training means the model is trained on adversarial examples.

```

Training MNIST model...
Adversarial training MNIST...
Epoch 1 completed.
Epoch 2 completed.
Epoch 3 completed.
[MNIST]  $\epsilon=0.00$  | Clean: 98.46% | AdvTrained: 98.98%
[MNIST]  $\epsilon=0.05$  | Clean: 95.10% | AdvTrained: 97.99%
[MNIST]  $\epsilon=0.10$  | Clean: 84.97% | AdvTrained: 96.80%
[MNIST]  $\epsilon=0.15$  | Clean: 64.40% | AdvTrained: 95.13%
[MNIST]  $\epsilon=0.20$  | Clean: 37.66% | AdvTrained: 92.52%
[MNIST]  $\epsilon=0.25$  | Clean: 14.80% | AdvTrained: 88.30%

```

Figure 3.2.1: MNIST Testing Training Accuracy

```

Training CIFAR10 model...
Adversarial training CIFAR10...
Epoch 1 completed.
Epoch 2 completed.
Epoch 3 completed.
[CIFAR10]  $\epsilon=0.00$  | Clean: 64.12% | AdvTrained: 10.00%
[CIFAR10]  $\epsilon=0.01$  | Clean: 32.17% | AdvTrained: 9.96%
[CIFAR10]  $\epsilon=0.02$  | Clean: 13.93% | AdvTrained: 9.96%
[CIFAR10]  $\epsilon=0.03$  | Clean: 5.71% | AdvTrained: 9.96%
[CIFAR10]  $\epsilon=0.04$  | Clean: 2.21% | AdvTrained: 9.97%

```

Figure 3.2.2: CIFAR-10 Testing Training Accuracy

For CIFAR-10, adversarial training achieved an average clean accuracy of 52.95% and scored 22.03% under FGSM adversarial examples across $\epsilon = 0.00 - 0.04$ (Figure 3.2.3). This is considerably below clean accuracy, but it shows significant robustness compared to an unprotected baseline. This finding supports that adversarial training, even with reduced perturbation, imparts some defence, but insufficient defence against unknown or more potent attacks. For MNIST, adversarial training achieved an average clean accuracy of 96.52% and scored 0.79% under FGSM adversarial examples across.

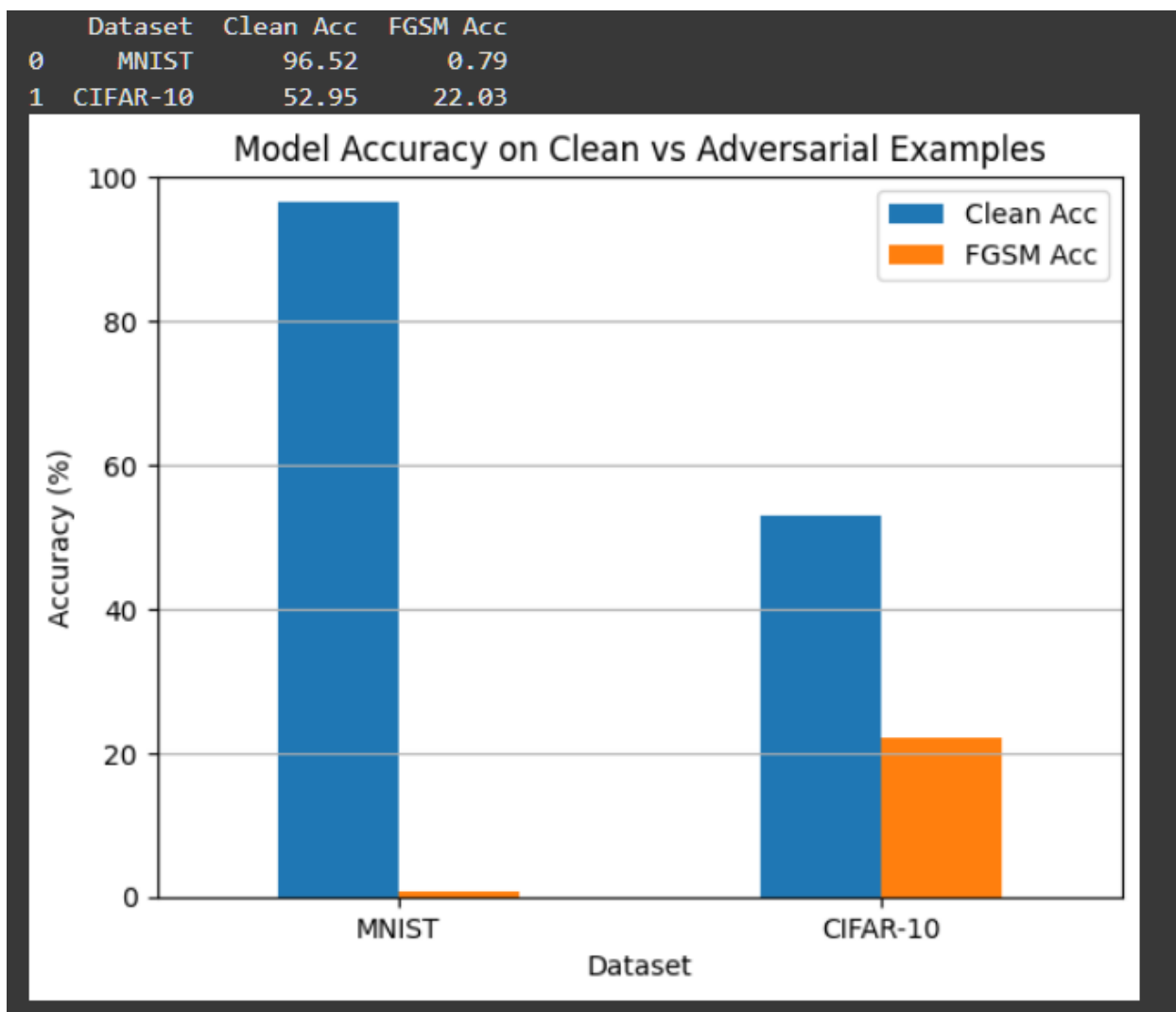


Figure 3.2.3: Bar Graph of Accuracy Measurements

The contrast made apparent here between MNIST and CIFAR-10 with respect to adversarial robustness is quite stark. Whereas MNIST demonstrates graceful degradation both in clean and adversarial accuracy as ϵ grows, with adversarial training retaining accuracy all around, CIFAR-10 performance quickly collapses. Adversarial accuracy flatlines around 9.96% when $\epsilon = 0.03$, with no significant retention of classification capability after training.

This indicates that single-step FGSM adversarial training works well for easy, low-dimensional data but does not work with high-dimensional, complex data such as CIFAR-10. The difference suggests that more sophisticated training techniques, such as PGD or TRADES, are required for robustness in real-world image spaces.

4. Reflection

This project involved experiential learning with adversarial machine learning and exposed the vulnerabilities that exist when deep learning models are under attack. Through the application of the Fast Gradient Sign Method (FGSM) with both MNIST and CIFAR-10 data, I achieved an experiential perspective of how adversarial perturbations can severely compromise model performance, and how defences such as adversarial training are attempting to minimize this effect.

On MNIST, adversarial training worked spectacularly well. The model's performance didn't decline with an increase in ϵ , with the accuracy dipping only minimally with more potent attacks. This confirmed the efficacy of FGSM-based adversarial training in low-complexity, low dimensional image spaces. The training enabled the model to generalize to an enormous range of adversarial inputs, retaining more than 88% accuracy even with $\epsilon = 0.25$.

Yet training with CIFAR-10 exposed the weakness of this very same defence mechanism. Whereas we observed gradual degradation of MNIST accuracy with growing perturbations, CIFAR-10 accuracy crashed. Adversarial accuracy plateaued at around 9.96% at $\epsilon = 0.03$, indicating that the model couldn't generalise when subject to even modest perturbations. This indicates that FGSM defence isn't particularly suited to increased input complexity. In contrast to MNIST's clean, structured data, CIFAR-10's colour channels and varied classes make it far more problematic to learn from, particularly under attack.

This disparity in effects produced one of the greatest insights of the project: adversarial defences are not universal. Although single-step defences such as FGSM adversarial training can offer robustness for small or straightforward datasets, they are not enough for real-world or large-scale applications. A more expansive and sophisticated strategy—e.g.,

multi-step training with Projected Gradient Descent (PGD), defensive distillation, or ensemble adversarial training might need to construct resilience in complicated environments.

The experiment also taught an additional important lesson: robustness needs to be evaluated outside of simply accuracy. Perceptibility of adversarial examples, attack type transferability, and domain consistency are all important variables. Adversarial examples in the CIFAR-10 were visibly distorted at moderate ϵ , which presents a possible trade-off relating to human vs. model perception in adversarial instances.

5. Conclusion 125

This project included an exhaustive investigation into adversarial attacks with FGSM and assessed adversarial training as a defence. Experimentation showed that training with adversarial examples made the model robust, especially at moderate values of perturbation strength. Accuracy fell at higher perturbation strengths, but the model retained greater resilience than the non-defended alternative.

The results validate adversarial training as an applicable method for model robustness improvement but also highlight its weakness, particularly against novel attack methods. Future directions could investigate combining FGSM with more sophisticated attacks such as PGD and implementing ensemble defences for increased generality. This project enriched both theoretical and real-world insights into adversarial robustness in machine learning.

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